

Recommended Assessment

LCD Display

Exploring LCD Display

1. When moving your finger around the screen, how many columns and rows did you think the display has? Where is (0,0) and where is (maxCol, maxRow)? Does the number of pixels you observed match what you see in the datasheet?
2. When using different materials to touch the screen, what did you notice? When does it read vs when does it not?
3. When printing the reads for multiple fingers, what happened to **finger ID** as you were removing and adding fingers? What is the maximum number of fingers you can read with this display? Does this match the datasheet?

Plotting Touch Input

4. What happens when you move your fingers fast vs slow when drawing on the screen? Note that the touch panel samples at 120Hz.
5. What happens when you are drawing with two fingers and you remove the first one that touched the screen? Is this what you would want to happen with your colors? How do you think you could fix this?
6. The display image is updated in the function control_loop(). The current image is stored in the array **image**, declared as a 3-dimensional array. What do each of the dimensions represent? In the following code, describe which elements are updated and explain why:

```
image[finger1.r:finger1.r+squareSize,  
finger1.c:finger1.c+squareSize,:] = colorSquare1
```

Why do we use **squareSize** here?

7. When plotting touch, explain how your horizontal and vertical lines appear when compared to the ones on your display. In the `update_plot` function, are the array elements defined in the same order that image array is updated? Why could the plots vary?
8. How did you modify **`update_plot`** to match the device screen to the plot? Show the picture you drew.

Writing to the Display

9. Based on what you saw in the draw modes file, what would be the difference between using **`draw_image_mask`** and **`draw_image_blend`** in the **`lcd_draw_touch.py`** file?

LCD Touch Display Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	